Research Program: Bala Krishnamoorthy

I lead an expansive and well-funded research program that is theoretically grounded on the themes of Geometry, Topology, Optimization (GTOpt), and their interactions. Built on the theoretical foundations are multiple application thrusts including additive manufacturing, computational chemistry, criminology, phenomics and agronomy, power grid optimization, astrophysics, as well as collaborations with national laboratories and industry.

I have raised over $19.68M in research funding from federal and state agencies (total amount, as PI or co-PI). I have been part of large team projects, e.g., the WADEPS (WA State Data Exchange for Public Safety, wadeps.org), in which the activities of diverse sets of stakeholders have to be coordinated before the data and optimization team headed by me can "do the Math". On the other hand, I have secured a sole PI grant from NSF on robust optimization for which I worked with my PhD and undergraduate students on research. But in most of my funded projects, I am collaborating with researchers from multiple disciplines outside of my own. I've built long term collaborations with national labs (Pacific Northwest National Laboratory, Air Force Research Laboratory, and Oak Ridge National Laboratory), which afford indirect funding through long-term research internships (8–12 months, or longer) for my PhD students to work on their dissertation research. Furthermore, I've developed a collaboration with Mitsubishi Electric Research Labs (MERL) on mathematical aspects of additive manufacturing (3D printing). Once again, a PhD student of mine is funded as an intern (13+ months), and the team (MERL researchers, my student, and I) have published papers as well as filed for non-provisional patents.

I co-founded a tech start-up (cognetrylabs.ai) that built AI-based headless commerce solutions for retailers. I headed the AI and Optimization division of the company and oversaw the intellectual property portfolio (I have a patent: US 12118604B2). My mathematical research is often informed by my work outside of academia.

On the theoretical side, I work on multiple problem domains from the GTOpt theme. The optimal homology problem (OHP) on a simplicial complex K seeks a cycle $\mathbf{x}$ that is homologous to a given cycle $\mathbf{c}$ (e.g., $\mathbf{c}$ and $\mathbf{x}$ both go around the same hole in K) such that some norm defined on $\mathbf{x}$ is optimized. While OHPs are NP-hard in general with homology defined over $\mathbb{Z}$ or $\mathbb{Z}_2$, I discovered a new topological property of K that is characterized by the total unimodularity of its boundary matrix that guarantees the solution of OHPs as linear programs (LPs). This connection has opened up the possibilities of building efficient network flow models for OHPs under specific settings. In related work, I have tackled fundamental (open for long time in some cases) problems from geometric measure theory (GMT), e.g., on integral flat norm decomposition of currents, using the simplicial complex + total unimodularity connection. Extending results connecting Steiner minimal trees and minimum spanning trees to the continuous setting in the limit, I have studied a version of the analyst's traveling salesman problem (TSP). In computational topology, I have developed new filtration frameworks using rectangular boxes (instead of default balls) that are more robust to noise as well as a Steinhaus filtration for abstract covers that have a measure defined on them. Motivated by time series applications in criminology, I have developed stable measures of comparison between time series as well as a measure to quantify chaos in dynamical systems using persistent homology. I have developed a new Euler mesh that guarantees a uniform even degree (4 in 2D, 6 in 3D) for all vertices, allowing the design of continuous toolpaths in 3D printing. I have developed theoretical convergence guarantees and provable bounds for scalable algorithms for stochastic optimization problems arising in the power grid, and have developed efficient algorithms for matching problems in the online setting.

On the applications side, I have worked on multiple problems in computational chemistry using tools from persistent homology, GMT, optimal transport, dimension reduction, and more. I've worked with agronomists on finding structure in large phenomics datasets using statistical machine learning and optimization. I have worked with large datasets from orthopedic surgeons and bioengineers. With criminologists, I am using tools from statistics, machine learning, and applied topology. I directly involve my students (PhD, MS, and undergraduate) in all such collaborations. In fact, each of my PhD students (15 graduated, 6 current) has worked on at least one applied project apart from their theoretical research. By training them in this multidisciplinary manner, I ensure my students become versatile problem solvers who are equally comfortable analyzing large datasets as they are in proving theorems.